

20/8/25

Task 3 - Importing and creating python modules and packages in python program

Aim:

To implement and demonstrate the process of importing built-in modules, creating user-defined modules and organizing code into packages in Python, thereby promoting code reusability, modularity and maintainability.

31 - Perform common math and random operations, works with the operating system and compute basic statistics.

Algorithm:

1. Import required modules; math, random, os, sys, statistics, pathlib

2. Math & random:

- Compute $\text{sqrt}(5)$, $\text{radians}(30)$, a random float in $[0, 0, 1.0)$, a random integer in $[2, 6]$ (inclusive), π , $\text{ceil}(2.3)$, $\text{floor}(2.3)$, $\text{factorial}(5)$, $\text{gcd}(5, 15)$, $\text{abs}(-10)$, $\text{pow}(3, 5)$, log base 3 of 2, $\text{log10}(e)$ for $a=100$, and check NaN/infinity.

3. OS & sys:

- List all files in the new current directory
- Print Python interpreter version

4. Statistics:

- On Lists: $[5, 6, 8, 10]$ and $[2, 5, 3, 2, 8, 3, 9, 4, 2, 5, 6]$, compute mean, median, mode, standard deviation

5. Print neatly for formatted results.

Program:

```
import math
import random
import os
import sys
import statistics as stats
from pathlib import Path
```



```

print ("\n --- MATH & RANDOM ---")
print ("sqrt(5) = ", math.sqrt(5))
print ("radians(30) = ", math.radians(30))
print ("random() in [0,1) = ", random.random())
print ("randint(2,6) = ", random.randint(2,6))
print ("pi = ", math.pi)
print ("ceil(2.3) = ", math.ceil(2.3))
print ("floor(2.3) = ", math.floor(2.3))
print ("factorial(5) = ", math.factorial(5))
print ("gcd(5,15) = ", math.gcd(5,15))
print ("abs(-10) = ", abs(-10))
print ("pow(3,5) = ", pow(3,5))
print ("log base 3 of 2 = ", math.log(2,3))
a_val = 100
print (f"log10 ({a_val}) = ", math.log10(a_val))
inf_val = float('inf')
nan_val = float('nan')
print (f"is inf (∞) = {math.isinf(inf_val)}, is nan (NaN) = {math.isnan(nan_val)}")
print ("\n --- OS & SYS ---")
path_python_lab = Path(r"C:\PythonLab")
path_python_lab.mkdir(parents=True, exist_ok=True)
print (f"Created/ensured : {path_python_lab}")
print ("Current working directory : ", os.getcwd())
target_dir = Path(r"C:\Python\SL4")
target_dir.mkdir(parents=True, exist_ok=True)
os.chdir(target_dir)
print (f"Changed into : {target_dir}")
print ("Directory contents : ", os.listdir())
print ("Python version = ", sys.version)

```

```

print("\n - - - STATISTICS - - ")
data1 = [5, 6, 8, 10]
data2 = [2, 5, 3, 2, 8, 3, 9, 4, 2, 9, 6]
print(f"mean({data1}) = ", stats.mean(data1))
print(f"median({data1}) = ", stats.median(data1))
print(f"mode({data2}) = ", stats.mode(data2))
print(f"stdev({data2}) = ", stats.stdev(data2))

```

3.2. Create a python package named Cardpack containing a module Cardfun that imports the random module. Assign a range of cards, call a function from the module, and display a random using sample of cards.

Algorithm:

Step: 1 - start

Step: 2 - To create a package Cardpack

Step: 3 - To create a module Cardfun and import random function

Step: 4 - Assigns a cards range

Step: 5 - Call a module functions

Step: 6 - Display the random sample cards.

Step: 7 - stop

Program:

Cardfun

import random

def func():

cards = []

for i in range(1, 53):

Expected Sample Output:

---MATH & RANDOM---

$\text{sqrt}(5) = 2.23606797749979$

$\text{radians}(30) = 0.5235987755982988$

$\text{random}() \text{ in } [0,1) = 0.37444887175646646$

$\text{randint}(2,6) = 6$

$\pi = 3.141592653589793$

$\text{ceil}(2.3) = 3$

$\text{floor}(2.3) = 2$

$\text{factorial}(5) = 120$

$\text{gcd}(5,15) = 5$

$\text{abs}(-10) = 10$

$\text{pow}(3,5) = 243$

$\log_{\text{base } 3 \text{ of } 2} = 0.6309297535714574$

$\log_{10}(100) \approx 2.0$

$\text{isint}(0) = \text{True}$, $\text{isnan}(NaN) = \text{True}$

-- OS & SYS --

Created / ensured : C:\Pythonlab

Current working directory : C:\... (your current path)

Created / ensured & changed into : C:\Python\slot 5244

Directory contents of C:\Python\slot 5244 : []

Python version : 3.x.x (... details...)

--- STATISTICS ---

$\text{mean}([5,6,8,10]) = 7.25$

$\text{median}([5,6,8,10]) = 7.0$

$\text{mode}([2,5,3,2,8,3,9,2,5,6]) = 2$

Output:

RESTART

C:\Users\Student.MAT2V C6833\AppData\Python\Python
Lib\site-packages\CardPack\My mod.py

[5, 24, 13, 22, 20, 41, 38, 51, 4, 7, 34, 49, 14, 50, 33, 40, 15,
17, 18, 33, 39, 36, 42, 12, 6, 16, 19, 48, 29, 2, 27, 11, 31, 46, 28,
32, 8, 25, 30, 23, 26, 10, 43, 47, 3, 44, 52, 1, 45, 9]


```
cards.append(i)
shuffled_cards = random.sample(cards, k=52)
print("\n\n", shuffled_cards, "\n\n")
```

My mod.py

```
import cardfun
(cardfun.fun1())
```

3.3. You are tasked with developing a modular calculator application in Python. The calculator should support basic arithmetic operations.

Algorithm:

1. Define functionally for addition, subtraction, multiplication and division
2. Handle division by zero by raising an error if the divisor is zero
3. Import the module (my math) contain these functions
4. Initialize two numbers ($a=10, b=5$). Call each function using `my math <function_name> (a,b)`.
5. Print the results of all operations

Program: (my) math

```
def add(a,b):
```

```
    return a+b
```

```
def subtract(a,b):
```

```
    return a-b
```

```
def multiply(a,b):
```

```
    return a*b
```

```
def divide(a,b):
```

```
    if (b==0):
```

```
        raise ValueError("cannot divide by zero")
```

Output:

Addition: 15

Subtraction: 5

Multiplication: 50

Division: 2.0

Calculator application in Python. The calculator should support basic arithmetic operations.

Algorithm:

1. Define functionally for addition, subtraction, multiplication and division.

2. Handle division by zero by raising an error in the division.

3. Import the module (operator) to use these functions.

4. Initialize two numbers (a=10, b=5). Call each function using numbers & function name > (a, b).

5. Print the results of all operations.

Program: (C++ code)

```
def add(a, b):
```

```
    return a + b
```

```
def subtract(a, b):
```

```
    return a - b
```

```
def multiply(a, b):
```

```
    return a * b
```

```
def divide(a, b):
```

```
    if b == 0:
```

ValueError: Cannot divide by zero


```
import mymath
```

```
a=10
```

```
b=5
```

```
print("Addition:", mymath.add(a,b))
```

```
print("Subtraction:", mymath.subtract(a,b))
```

```
print("Multiplication:", mymath.multiply(a,b))
```

```
print("Division:", mymath.divide(a,b))
```

3.4 Work on a python project that requires you to perform various mathematical operations and geometrical area calculations. Demonstrate the use of the package in functions by performing a few calculations and printing the results.

Algorithm:

1. Create math functions.py module.

2. Create area functions.py module

3. Create main.py:

4. Print the output as expected

Program:

Mathfunctions.py

```
def Add(a,b):  
    return a+b
```

```
def Sub(a,b):  
    return a-b
```

```
def mul(a,b):  
    return a*b
```

```
def div(a,b):
```

```
    if b==0:
```

```
        return "Error! Division by zero"
```

```
    return a/b
```


Areafunctions.py

```
import math
```

```
def circle-area(radius):
```

```
    return math.pi * radius * radius
```

```
def rectangle-area(length,width):
```

```
    return length * width
```

```
def triangle-area (base,height):
```

```
    return 0.5 * base * height
```

main.py

```
import math functions
```

```
import Areafunctions
```

```
print ("Addition:", math functions .Add(10,5))
```

```
print ("Subtraction:", math functions .sub(10,5))
```

```
print ("Multiplication:", math functions .mul(10,5))
```

```
print ("Division:", math functions .div(10,5))
```

```
print ("Circle Area (radius=7):", area functions .circle-area
```

(1))

```
print ("Rectangle Area (5x10):", area functions .rectangle-area
```

(5,10))

```
print ("Triangle Area (base=6,height=8):", area functions .triangle-
```

area(6,8))

VEL TECH	
EX NO.	3
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	
DATE WITH DATE	15

Result:

Thus, the program for importing Python modules and Packages was successfully executed and the output was verified.

Output:

Addition = 15

Subtraction = 5

Multiplication = 50

Division = 2.0

Circle Area (radius = 7): 153.93804002589985

Rectangle Area (5 x 10) = 50

Triangle Area (base = 6, height = 8) = 24.0